

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Provisioned and operated **multi-cloud GPU clusters** (AWS, GCP, CoreWeave) with **NVIDIA/CUDA** hardware and **InfiniBand/RoCE** networking for long-running training and **high-availability inference**.
- Ran multi-GPU **PyTorch** distributed training via **Slurm** HPC scheduling; built **Prometheus/Grafana** observability and **autoscaling** for **vLLM** serving, keeping fleet utilization up through **node loss**.
- Wrote **Go** and **C++** for training-job **scheduling**, **resource allocation**, and **fault-tolerant placement** under GPU scarcity.
- Implemented fault-tolerant recovery for **Python** pipelines via per-node **Redis checkpointing**, resuming interrupted workflows from the last committed step; deployed to **EKS** at **99%+ uptime**.
- Built real-time **observability** with **automated incident detection**, **rollback guardrails**, and **runbooks** for production pipelines.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Built **Airflow DAGs** on **GCP** orchestrating **PySpark** jobs for large-scale dataset preparation feeding **model training** and **evaluation**, with automated **failure alerting**.
- Cut **p99 latency 420ms to 140ms** across **50K+ daily requests** on **GCP** with **Kubernetes-managed autoscaling**; built automated **CI/CD rollback**, alerting, and **on-call incident triage**.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Built an automated **model-evaluation and deployment-gating framework** covering quality, latency, robustness, and regression checks with rollback criteria; cut model-related production regressions **28%** and validation time from **6 hours to 90 minutes**.
- Profiled compute/memory bottlenecks and optimized **batching**, preprocessing, tensor movement, and request scheduling across **Python/C++** inference services, increasing throughput **2.4x** while cutting GPU memory usage **27%**.
- Operated **25+ production microservices (C++, SQL)** for **1M+ users** at **99.9% uptime** on **AWS**.
- Standardized API contracts and failure handling across **25+ services**, reducing escalations **60%**; drove **root cause analysis** and **post-incident reviews** to improve long-term reliability.

PROJECTS

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- Raft-consensus** KV store with **linearizable reads** and **log compaction**; companion **VPC control plane** simulator.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- LangGraph** multi-agent system triaging **production incidents**; synthesizes **root cause reports**, reducing **MTTR**.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Production **observability** platform with **regression detection**, **alerting**, and latency tracking; **FastAPI**.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages & Fundamentals: Python, C++, Go, Java, SQL; Data Structures and Algorithms, Distributed Systems

ML Infra & GPU: PyTorch, Multi-GPU Distributed Training, NVIDIA/CUDA, InfiniBand/RoCE, Slurm, vLLM, TensorRT-LLM

Cloud & Orchestration: Kubernetes, Docker, GCP (Cloud Run, BigQuery, Pub/Sub), AWS (EKS, S3, Lambda), Terraform

Data & Observability: Airflow, PySpark, Kafka, Redis, PostgreSQL, Prometheus, Grafana, Alerting, On-call, Runbooks

CERTIFICATIONS

AWS Certified Developer Associate | GCP Professional MLOps Engineer